

ASSIGNMENT-1

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COURSE : CRYPTOGRAPHY

1) Question-1 Caesar cipher

a) Encrypt "DEFEND THE WALL" with Key = 3

Shift each letter by +3.

Plaintext

DEFEND THE WALL

Cipher text

GHIHQG KHW ZDDD

95
100
5
20/100

b) Decrypt "KHOOR ZRUOG" with Key = 3

Shift each letter by -3

Cipher text

KHOOR ZRUOG

Plaintext

Hello world

c) SECRET → VHIFUHW

Compare letters : S → V (+3)

Therefore,

Key = 3

Decrypt

WKLV LV O WHWK

Plaintext

THIS IS A TEST

2) Question 2 - Monoalphabetic Substitution Cipher:

a) Encrypt "FIRE"

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Q	W	E	R	T	Y	V	I	O	P	A	S	D	F	G	H	J	H	L

T	U	V	W	X	Y	Z
Z	X	C	V	B	N	M

Encryption:

Cypher text

F → Y

I → O

R → K

E → T

Y O K T

b) Decrypt:

Cypher text

UJHHD QEB NRFZH YDCTH

Plain text:

THE QUICK BROWN FOX

Approximate english sentence obtained using frequency analysis)

3) Question 3 - Vigenere cipher

Keyword:

LEMON

a) Encrypt:

plaintext

- ATTACK AT PALACE

Repeat keyword

LEMONLEMONLEMO

Ciphertext

LXFDPVERNHRMH

b) Decrypt

ciphertext

LXFDPVERNHR

Keyword:

LEMON

Plaintext

ATTACK AT DAWN

c) Decrypt

ciphertext

QNXEPKMACTU

Keyword:

key

Plaintext

GOTALACWAPK

4) Question 4 - Hill cipher.

Use matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

a) Encrypt "HI"

$$H = 7$$

$$I = 8$$

vector

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

multiply

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

modulo 26

$$45 \bmod 26 = 19 = T$$

$$54 \bmod 26 = 2 = C$$

Ciphertext

TC

b) Verify "PO → QC"

$$P = 15$$

$$O = 14$$

multiply

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 15 \\ 14 \end{bmatrix}$$

$$= \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

modulo 26

$$87 \bmod 26 = 9 = J$$

$$100 \bmod 26 = 22 = W$$

Result

JW

5) Question 5 - playfair cipher.

Keyword:

ZEBRAS

a) Encrypt.

plaintext

MEETMEATERLUNCH

Ciphertext

NFKUNFBDSWPDQUGK

Keyword

BOARD

b) Decrypt:

ciphertext

EANNMTEFLUHEIREL

plaintext

MEET MEATERNDON

6) Question 6 - One-Time pad.

a) Encrypt.

plaintext

Hello

key

XMCKL

Convert to number

H = 7	X = 23
E = 4	M = 12
L = 11	C = 2
L = 11	K = 10
O = 14	L = 11

Add modulo 26

$$7 + 23 = 30 \rightarrow A = E$$

$$4 + 12 = 16 = Q$$

$$11 + 2 = 13 = N$$

$$11 + 10 = 21 = V$$

$$14 + 11 = 25 = Z$$

Ciphertext

EQNVZ

b) Decrypt

b) Why OTP is unbreakable:

Key length equals message length

Key is completely random

Key is used only once

Key remain secret

Every plaintext is equally likely

7) Question 7 - Columnar Transposition.

a) Encrypt.

Plaintext

WEARE DISCOVERED

Keyword

ZEBRA

Arrange:

Column order

Cipher Text

ZEBRA

WEARE

DISCO

VERED

A B E R Z

5 3 2 1 1

EOOASREICEDWD

b) Decrypt

Ciphertext

EVLNEACOTKESEAQRFECOKWN

keyword

TRIAL

Recovered plaintext

WE ARE DISCOVERED FLEE AT ONCE

c) Encrypt

plaintext

INFORMATION SECURITY

Keyword

HACK

length = 19

column = 4

Row

$19 \div 4 = 4$ remainder 3

Rows needed = 5

padding x

Ciphertext

O T E I I F O T R R M A Y N N S C U T I

b) Decrypt

Ciphertext

EVLNEACOTKESEARDFECOKWN

Keyword

TRIAL

Recovered plaintext

WE ARE DISCOVERED FLEE AT ONCE

c) Encrypt

Plaintext

INFORMATION SECURITY

Keyword

HACK

Length = 19

Column = 4

Row

$19 \div 4 = 4$ remainder 3

Rows needed = 5

padding x

Ciphertext

~~O T E I I F O T R R M A Y N N S C O U I~~

Question 8 - Frequency Analysis.

Ciphertext

UZ QSOVDHX MOPVCIPEUSGIWZOPFPESXZOTUCCG

Frequency

Z → 8

O → 7

U → 5

P → 5

S → 4

E → 4

Probable mapping

Z → F

O → T

P → A

U → D

S → I

Partial plaintext

DE ... TO ...

9) Question 9 - known plaintext Attack.

Ciphertext

BJJW QJW YMJW NX F XNRQJWX

known plaintext

This is The

mappings:

B $\rightarrow$ T

J $\rightarrow$ H

W $\rightarrow$ I

Q $\rightarrow$ S

Y $\rightarrow$ E

M $\rightarrow$ T

N $\rightarrow$ I

X $\rightarrow$ S

Recovered plaintext

This is the a Simple

~~Final message~~

✓ This is the Simple text

10) Question 10 - Vigenere Cipher (unknown key)

Ciphertext

ZICVTWQNGRZGVTWAVZHCQYCLMGU

Probable key

DECEPTIVE

Decrypted plaintext

WE ARE DISCOVERED SAVE YOURSELF

Final Answer.

Recovered key: DECEPTIVE

Plaintext: WE ARE DISCOVERED SAVE
YOURSELF.

D → E

H → L

I → W

E → D

F → V

P → M

I → U

E → S